Prem Patel

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Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science | GPA: 3.8 — *Expected Dec. 2026*

Wilfrid Laurier University — Waterloo, ON
Bachelor of Business Administration | President's Gold Scholarship — *Expected Dec. 2026*

Experience

Cloudflare — Sep. 2026 – Present
Software Engineer Intern — *Austin, TX*

- Engineering a **distributed latency-probing service (TypeScript)**, coordinating **HTTP/TCP** measurements across **200+ edge data centers** to optimize Cloudflare Worker placement, replacing a legacy bare-metal service
- Developed namespace-scoped **role-based access control** in **Go** for **Workers for Platforms**, extending backend APIs and collaborating with IAM engineers to design hierarchical authorization for secure enterprise deployments

Tesla — May 2026 – Aug. 2026
Software Engineer Intern — *Palo Alto, CA*

- Built a **scalable analytics backend** for Powerwall residential batteries enrolled in **Virtual Power Plant** programs **(Scala, Rust)**, reducing response time for VPP partner escalations by **90%**
- Iterated on an **ETL prototype** with user feedback, then **re-architected** and built a **production system** with a **gRPC API**, indexed **PostgreSQL** schema, **Spark**-backed data-writing layer, and client auth/access controls
- Developed distributed **Scala services** to aggregate and process performance data from **millions of Powerwall IoT devices**, deploying through **Argo CD** and **Kubernetes**

Kinaxis — Jan. 2025 – Apr. 2026
Software Engineer Intern — *Ottawa, ON*

- Built end-to-end features in a **distributed C++ server** with **Java** + **TypeScript (Node.js)** agents/workers over persistent **WebSockets**, improving reliability, latency, and throughput under load
- Shipped **BYO-Model** with a solver-agnostic, **async job pipeline** to run optimization models at scale
- Designed thread-safe CPU/memory **guardrails** in **C++** using **RAII semaphores** to prevent noisy-neighbor resource spikes in **multi-tenant** clusters; cut lock contention by **32%**
- Implemented an **IPC state handler** with retry scheduling and **disconnection detection**, improving **fault tolerance** and preventing cascading failure logs after worker/agent disconnects
- Improved response serialization with **FlatBuffers** to support **25%** larger payloads and eliminated silent **Node.js** crashes, increasing throughput and reliability under load
- Standardized **structured logging** + metrics integrated with Datadog dashboards to attribute CPU usage per customer, enabling transparent **consumption-based pricing** supporting **$1M+** annual revenue
- Deployed EA services on **Kubernetes** via **Docker** + **Helm** (**GCP**) and presented **17** cross-team demos

AutoTrader — Jan. 2024 – May 2024
Software Engineer Intern — *Toronto, ON*

- Developed **Python** ingestion services processing OEM feeds for **30+ automotive brands**; reduced end-to-end processing time by **8x** by restructuring transformations and batching (Pandas + REST API integrations)
- Built **Azure Pipelines** automation to generate and deliver monthly reports for **150+ dealerships**, integrating **Redash** queries and enforcing consistent schemas/validation

Technical Skills

Languages: Java, Python, Go, TypeScript (JavaScript), C++, Scala, Rust, C#, SQL, Bash
Frameworks/Libs: gRPC, Spring Boot (DI, WebSockets), Apache Spark, React, FastAPI, Boost, FlatBuffers
Dev Tools: Git, Node.js, Docker, Kubernetes, Helm, Argo CD, PostgreSQL, Jenkins, Datadog, GCP, Azure, AWS